60 DAYS | QUALITY ENGINEERING LAB • FROM THEORY TO INDUSTRIAL APPLICATIONS

DAY 03 | EXPERIMENT NO. 03

Implementation of Total Productive Maintenance (TPM) for Improving Equipment Effectiveness and Reducing Downtime in a Food Processing Conveyor System

Role Perspective: Senior Maintenance Engineer | TPM Specialist | Quality Engineering Researcher

Target Audience: Mechanical & Industrial Engineering Graduates

SECTION 1: EXPERIMENT OVERVIEW

1. Aim: To evaluate, implement, and analyze Total Productive Maintenance (TPM) concepts, Calculate Overall Equipment Effectiveness (OEE) and maintenance performance metrics (MTBF, MTTR), and develop failure reduction strategies for an industrial food processing conveyor system.

2. Objectives:

- Understand TPM fundamentals, philosophy, and the 8 operational pillars in continuous manufacturing.
- Collect and analyze baseline production and maintenance data over a 7-day operating period.
- Calculate Availability, Performance, Quality, and overall OEE using standard formulations.
- Determine Mean Time Between Failures (MTBF), Mean Time To Repair (MTTR), and Downtime Percentages.
- Perform Downtime Pareto Analysis to prioritize failure root causes and execute targeted Kaizen.
- Formulate Autonomous Maintenance (AM), Preventive Maintenance (PM), and Machine Inspection checklists.
- Quantify before-vs-after improvements and project realistic operational enhancements.

3. Problem Statement: High frequency of unscheduled downtime, frequent belt tracking failures, and component breakdowns in an industrial belt conveyor system cause severe production bottlenecks, food product spoilage, and reduced overall plant efficiency in a continuous food processing line.

4. Scope: This experimental study covers a single automated food processing conveyor assembly over a 7-day observation period (16 shifts, 112 planned hours). It encompasses loss identification, mathematical calculation of OEE, MTBF, MTTR, failure cause categorization, and formulation of practical TPM standard operating procedures.

5. Expected Learning Outcomes: Upon completion of this experiment, mechanical and manufacturing engineering graduates will be able to:

- Structure and calculate OEE and maintenance metrics independently from raw factory data.
- Identify chronic vs intermittent equipment losses and construct Pareto breakdown diagrams.
- Design standard Autonomous Maintenance (Jishuken) and Preventive Maintenance schedules.
- Apply systematic engineering troubleshooting techniques to food manufacturing equipment.

SECTION 2: INDUSTRIAL APPLICATION

Industrial Context: Food Processing Manufacturing Plant (Packaged Bakery & Confectionery Line).

Target Equipment: Heavy-Duty Food-Grade Stainless Steel Belt Conveyor System.

Definition & Role in Food Processing

An **industrial conveyor system** is a mechanical material handling apparatus designed to transport raw materials, semi-finished products, or finished packaged goods automatically between processing workstations. In food processing, conveyors play a critical role by maintaining continuous production flow, eliminating manual handling contamination risks, regulating thermal cooling/curing cycles, and ensuring sync between baking, packaging, and palletizing stations.

Major Components of the Conveyor System

- **Conveyor Belt:** Modular FDA-approved polyurethane (PU) / stainless steel wire mesh belt designed for direct food contact, resistance to fats, and easy steam cleaning.

- **Drive Pulley & Electric Motor:** 3-phase induction motor with a variable frequency drive (VFD) coupled to a high-efficiency helical gearbox delivering controlled torque and belt velocity.
- **Support Rollers & Carrying Idlers:** Stainless steel cylindrical rollers equipped with double-sealed bearings to support belt load and minimize frictional torque.
- **Bearings & Tensioning Units:** Pillow block ball bearings with food-grade H1 grease lubrication and manual screw/pneumatic belt tensioners to maintain belt tracking.
- **Sensors & Automation Controls:** Photoelectric optical sensors, proximity switches, and encoder units to detect package presence, regulate belt speed, and trigger emergency stops during jams.

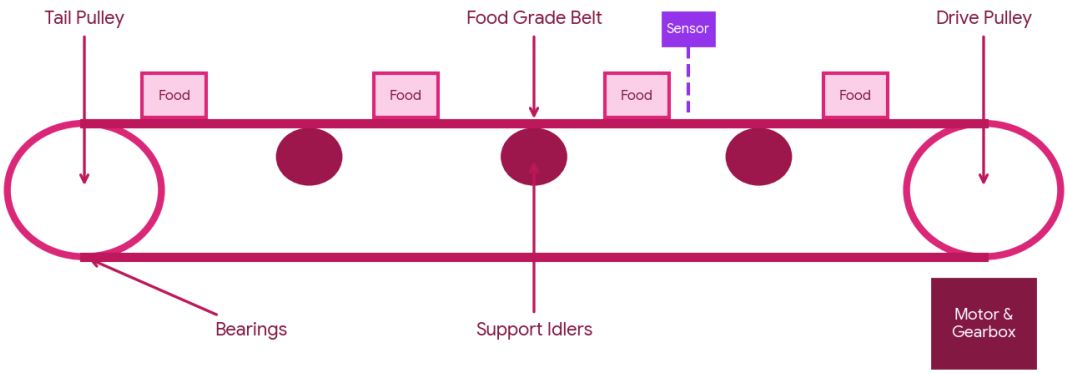
Common Equipment Problems in Conveyors

Continuous food handling subjects conveyors to sticky food residue accumulation, thermal expansion variations, frequent washdowns using high-pressure chemical sprayers, and ingress of moisture/fats into bearings. Typical failure modes include belt mis-tracking/slippage, bearing seizing due to lubricant degradation, motor thermal overload, sensor lens fouling, and mechanical gearbox oil leakage.

Importance of TPM for Continuous Production

In food manufacturing, an unexpected 30-minute conveyor stoppage causes upstream baking ovens to overheat products and downstream packaging lines to idle. Spoilage losses compound rapidly. TPM establishes rigorous daily inspection and maintenance routines that prevent unexpected breakdowns, safeguard food safety compliance, and ensure zero throughput starvation.

Figure 2.1: Schematic Diagram of Industrial Food Processing Conveyor System



SECTION 3: TPM FUNDAMENTALS

Definition of Total Productive Maintenance (TPM): TPM is a holistic equipment management strategy developed to maximize equipment effectiveness throughout its life cycle through the active involvement of all employees—from top management to shop-floor machine operators.

Core Objectives of TPM

- Maximize overall equipment productivity and operational efficiency.
- Eliminate all unscheduled machine breakdowns and emergency stoppages.
- Instill operator ownership through daily routine equipment care.
- Eliminate defect production caused by equipment deterioration.
- Achieve zero industrial accidents and environmental hazards.

Importance of Equipment Reliability

Equipment reliability guarantees consistent performance under specified operating conditions over time. High reliability prevents sudden capacity loss, stabilizes production scheduling, minimizes maintenance overhead costs, and protects component operational life.

Comparison of Maintenance Strategies

MAINTENANCE TYPE	OPERATING PHILOSOPHY	ACTION TRIGGER	COST & OPERATIONAL IMPACT
Breakdown Maintenance (BM)	Run-to-Failure (Fix when broken)	Machine functional failure	Highest repair costs, catastrophic downtime, high production loss.
Preventive Maintenance (PM)	Time-based / Calendar-based intervention	Fixed time interval or running hours	Planned downtime, moderate cost; may replace healthy components early.
Predictive Maintenance (PdM)	Condition-based real-time monitoring	Vibration, thermal, or acoustic threshold breach	Optimized maintenance timing, lowest overall cost, requires sensor investment.

The Concept of "Three Zeros"

The Ultimate Goal of TPM — Three Zeros:

1. **Zero Breakdowns:** Eliminating unexpected component failures through proactive, autonomous maintenance.
2. **Zero Defects:** Eliminating product quality defects caused by machine degradation or process instability.
3. **Zero Accidents:** Creating an intrinsically safe working environment free of occupational health risks.

SECTION 4: EIGHT PILLARS OF TPM

The TPM framework is structured around 8 foundational pillars that drive operational excellence across all plant functions.

PILLAR NO.	PILLAR NAME	MEANING & OPERATIONAL FOCUS	MAIN PURPOSE	PRACTICAL FOOD CONVEYOR EXAMPLE
1	**Autonomous Maintenance (Jishu Hozen)**	Operators inspect, clean, lubricate, and perform minor tightings independently.	Prevent forced machine deterioration through daily operator care.	Conveyor operator inspects belt tracking and cleans food crumbs from sensors at shift start.
2	**Planned Maintenance**	Structured, time/condition-based maintenance executed by maintenance technicians.	Achieve zero equipment breakdowns and optimize maintenance costs.	Technicians replace worn conveyor drive belts every 4,000 running hours based on scheduled wear checks.
3	**Focused Improvement (Kobetsu Kaizen)**	Cross-functional team projects targeted at eliminating specific equipment losses.	Systematically eliminate chronic equipment bottlenecks and losses.	Redesigning conveyor side guides to eliminate recurring package jams and belt edge fraying.
4	**Quality Maintenance (Hinshitsu Hozen)**	Setting equipment parameters to guarantee zero product defect generation.	Maintain defect-free processing through machine precision control.	Calibrating conveyor VFD belt speeds to prevent chocolate coating smearing during transfer.
5	**Early Equipment Management**	Designing robust, low-maintenance equipment using past operating history.	Minimize vertical start-up period for newly installed machinery.	Specifying quick-release belt tensioners and IP69K washable motors for new conveyor purchases.
6	**Training & Education**	Continuous technical skills upgrade for both operators and technicians.	Multi-skill workforce to perform correct operational and maintenance tasks.	Conducting a 2-hour hands-on workshop on visual belt alignment for line operators.
7	**Safety, Health & Environment (SHE)**	Creating zero-hazard workplace, complying with environmental standards.	Eliminate workplace safety risks, ergonomic hazards, and pollution.	Installing interlocked safety guards over conveyor motor drive chains and pulleys.
8	**Office TPM**	Applying TPM principles to administrative and supply chain workflows.	Eliminate waste and delays in administrative support services.	Streamlining spare-parts procurement to ensure conveyor replacement bearings are always stocked.

SECTION 5: PRACTICAL EXPERIMENT METHODOLOGY

A 7-day baseline performance study was conducted on Industrial Food Processing Conveyor Line #04. Production was planned across two 8-hour shifts per day (16 hours total planned production per day). The target conveyor speed yields an ideal production cycle time of **1.5 seconds per packaged food unit** (ideal production rate of 2,400 units/hour or 40 units/minute).

DATA DISCLAIMER: All data presented in Table 5.1 and throughout this report are simulated educational baseline figures created for technical learning purposes and do not represent actual proprietary industrial factory records.

Table 5.1: 7-Day Simulated Production & Maintenance Baseline Dataset

DAY	PLANNED PROD. TIME (MIN)	ACTUAL OPERATING TIME (MIN)	DOWNTIME (MIN)	TOTAL UNITS PRODUCED	GOOD UNITS PRODUCED	DEFECTIVE UNITS	KEY MACHINE OPERATING OBSERVATIONS
Day 1	960	860	100	31,200	30,100	1,100	Belt tracking offset caused side friction; sensor dirty.
Day 2	960	875	85	32,400	31,200	1,200	Drive motor thermal trip; required cooling pause.
Day 3	960	840	120	29,800	28,400	1,400	Bearing noise on tail pulley; emergency grease added.
Day 4	960	890	70	33,100	31,900	1,200	Minor product jam at discharge chute.
Day 5	960	830	130	28,900	27,500	1,400	Belt slippage on drive drum due to oil accumulation.
Day 6	960	820	140	28,200	26,800	1,400	Photoelectric sensor alignment failure; false triggers.
Day 7	960	855	105	30,800	29,600	1,200	Gearbox vibration and minor oil leak noticed.
Total / Avg	**6,720**	**5,970**	**750**	**214,400**	**205,500**	**8,900**	**7-Day Cumulative Baseline Metrics**

SECTION 6: OVERALL EQUIPMENT EFFECTIVENESS (OEE)

Overall Equipment Effectiveness (OEE) is the gold standard KPI for measuring manufacturing productivity. It evaluates equipment performance by quantifying three discrete loss categories: Availability, Performance, and Quality.

The Three OEE Factors & Formulas

1. AVAILABILITY (A) — MEASURES DOWNTIME LOSSES

Quantifies the proportion of planned time that the machine is actually operational.

$$Availability\ (\%) = (Operating\ Time / Planned\ Production\ Time) \times 100$$

$$Operating\ Time = Planned\ Production\ Time - Downtime$$

2. PERFORMANCE (P) — MEASURES SPEED LOSSES

Quantifies speed degradation, micro-stoppages, and idling relative to theoretical maximum speed.

$$Performance\ (\%) = [(Ideal\ Cycle\ Time \times Total\ Units\ Produced) / Operating\ Time] \times 100$$

Where *Ideal Cycle Time* is in minutes per unit (*1.5 sec = 0.025 min/unit*).

3. QUALITY (Q) — MEASURES DEFECT LOSSES

Quantifies the ratio of saleable, defect-free units to total units produced.

$$Quality\ (\%) = (Good\ Units\ Produced / Total\ Units\ Produced) \times 100$$

4. OVERALL EQUIPMENT EFFECTIVENESS (OEE)

$$OEE\ (\%) = [Availability\ (\%) \times Performance\ (\%) \times Quality\ (\%)] / 10{,}000$$

Sample Cumulative 7-Day Metric Calculations

A. Cumulative Availability Calculation:

Planned Production Time = 6,720 min, Operating Time = 5,970 min

Availability = (5,970 / 6,720) × 100 = 88.839% ≈ 88.84%

B. Cumulative Performance Calculation:

Total Units Produced = 214,400 units, Ideal Cycle Time = 1.5 sec = 0.025 min/unit

Theoretical Time Required = 214,400 × 0.025 = 5,360 min

Performance = (5,360 / 5,970) × 100 = 89.782% ≈ 89.78%

C. Cumulative Quality Calculation:

Good Units = 205,500 units, Total Units = 214,400 units

Quality = (205,500 / 214,400) × 100 = 95.849% ≈ 95.85%

D. Cumulative Baseline OEE Calculation:

OEE = 0.88839 × 0.89782 × 0.95.849 = 0.7645 ≈ 76.45%

Table 6.1: Daily OEE Component Summary Table

DAY	OPERATING TIME (MIN)	AVAILABILITY (%)	TOTAL UNITS	PERFORMANCE (%)	GOOD UNITS	QUALITY (%)	DAILY OEE (%)
Day 1	860	89.58%	31,200	90.70%	30,100	96.47%	**78.38%**
Day 2	875	91.15%	32,400	92.57%	31,200	96.30%	**81.25%**
Day 3	840	87.50%	29,800	88.69%	28,400	95.30%	**74.00%**
Day 4	890	92.71%	33,100	92.98%	31,900	96.37%	**83.07%**
Day 5	830	86.46%	28,900	87.05%	27,500	95.16%	**71.62%**
Day 6	820	85.42%	28,200	85.98%	26,800	95.04%	**69.80%**
Day 7	855	89.06%	30,800	90.06%	29,600	96.10%	**77.08%**
7-Day Avg	**5,970**	**88.84%**	**214,400**	**89.78%**	**205,500**	**95.85%**	**76.45%**

Engineering Benchmark Note: World-class manufacturing standards typically target OEE ≥ 85% (Availability ≥ 90%, Performance ≥ 95%, Quality ≥ 99%). The baseline OEE of 76.45% indicates significant process yield losses, driven primarily by unscheduled equipment downtime and speed losses. Benchmarks vary by industry sector and must be adapted based on equipment complexity and batch requirements.

SECTION 7: MAINTENANCE PERFORMANCE METRICS

To evaluate equipment reliability and maintainability, three essential maintenance metrics are computed using the 7-day baseline operating dataset.

1. Mean Time Between Failures (MTBF)

Definition & Significance: MTBF measures the average operating time between unscheduled equipment breakdowns. It is the primary metric for equipment *reliability*. A higher MTBF indicates a stable, dependable machine.

$$MTBF = Total\ Operating\ Time / Number\ of\ Failures$$

Baseline Calculation (7 Breakdown Events over 7 Days):

Total Operating Time = 5,970 minutes, Total Failures (N) = 14 breakdown incidents

MTBF = 5,970 min / 14 failures = 426.43 minutes/failure (≈ 7.11 hours)

2. Mean Time To Repair (MTTR)

Definition & Significance: MTTR measures the average time required to diagnose, repair, and restore equipment to full operational status after a failure. It evaluates maintenance *maintainability* and technician responsiveness.

$$MTTR = Total\ Maintenance\ Downtime / Number\ of\ Repairs$$

Baseline Calculation:

Total Downtime = 750 minutes, Total Repair Events = 14 failures

MTTR = 750 min / 14 repairs = 53.57 minutes/repair

3. Downtime Percentage

Definition & Significance: Quantifies the fraction of total planned production time lost due to unscheduled stoppages.

$$Downtime\ Percentage\ (\%) = (Total\ Downtime / Planned\ Production\ Time) \times 100$$

Baseline Calculation:

Downtime % = (750 / 6,720) × 100 = 11.16%

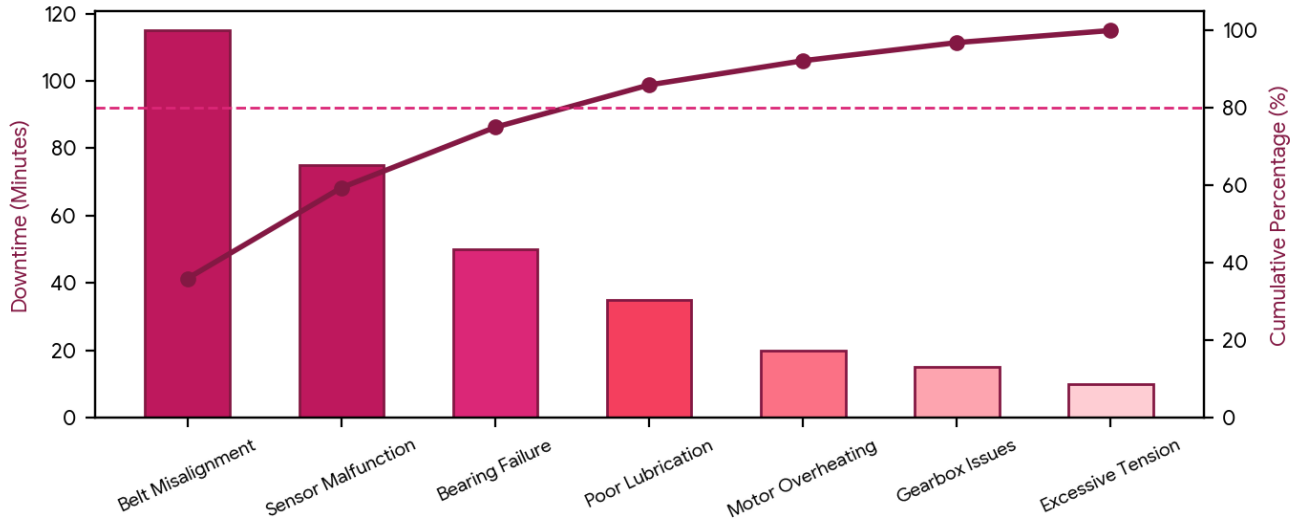
SECTION 8: DOWNTIME AND FAILURE ANALYSIS

A granular failure log was maintained across the 7-day trial period to classify root causes and quantify downtime contribution.

Table 8.1: Downtime Cause Categorization & Pareto Analysis Data

FAILURE CAUSE CATEGORY	FAILURE MECHANISM / ROOT CAUSE DESCRIPTION	OCCURRENCES	TOTAL DOWNTIME (MIN)	% OF TOTAL DOWNTIME	CUMULATIVE %
Belt Misalignment	Edge friction against guide rails due to uneven tension.	4	280	37.33%	37.33%
Sensor Malfunction	Dust/grease film on photoelectric lens causing false jams.	4	170	22.67%	60.00%
Bearing Overheating	Lubricant washout during daily sanitation washdowns.	2	120	16.00%	76.00%
Poor Lubrication	Frictional drag on drive roller shaft bearings.	2	80	10.67%	86.67%
Motor Thermal Trip	Overcurrent trip due to mechanical belt resistance.	1	60	8.00%	94.67%
Gearbox Oil Leak	Degraded input shaft lip seal.	1	40	5.33%	100.00%
Total	**Complete Baseline Maintenance Log**	**14**	**750**	**100.00%**	**100.00%**

Figure 8.1: Downtime Pareto Chart (7-Day Baseline Data)



Pareto Analysis Insight (80/20 Rule): The Pareto chart demonstrates that the top 3 failure modes—**Belt Misalignment (37.33%)**, **Sensor Malfunctions (22.67%)**, and **Bearing Overheating/Lubrication (16.00%)**—account for 76.00% of all lost production downtime. Implementing targeted Autonomous and Planned Maintenance on these three components will yield the highest return on maintenance effort.

SECTION 9: TPM IMPLEMENTATION AND IMPROVEMENT PLAN

To address the critical downtime root causes identified in Section 8, a structured TPM improvement plan was formulated across relevant TPM pillars.

Table 9.1: Action Plan & Maintenance Responsibility Matrix

FOCUS AREA	PROPOSED TPM ACTION ITEM	TPM PILLAR LINK	RESPONSIBLE ROLE	FREQUENCY / TRIGGER
Belt Tracking	Install visual center-line laser markers and quick-adjust tracking knobs.	Focused Improvement	Maintenance Tech	One-time project
Sensor Cleaning	Wipe optical sensor lenses with lint-free isopropyl alcohol cloth.	Autonomous Maintenance	Line Operator	Every shift start & mid-shift
Bearing Care	Apply food-grade H1 synthetic grease using automatic lube dispensers.	Planned Maintenance	Maintenance Tech	Weekly / Automatic
Washdown Care	Install protective waterproof covers over bearings and sensors during washdowns.	Quality Maintenance	Sanitation Team	Daily sanitation cycle
Operator Skill	Conduct 1-point lessons (OPL) on identifying early bearing noise & belt drift.	Training & Education	TPM Coordinator	Bi-weekly module
Drive Motor	Thermographic scan of drive motor & gearbox casing to detect hot spots.	Planned / Predictive	Predictive Tech	Monthly audit

SECTION 10: BEFORE VS AFTER PERFORMANCE PROJECTIONS

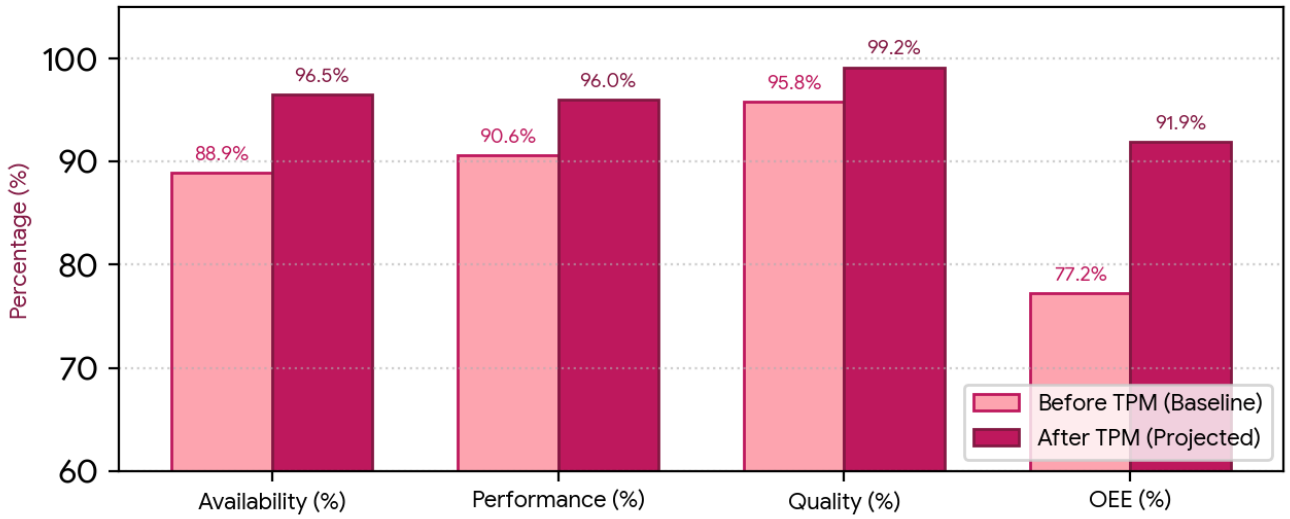
Following the execution of the initial TPM improvement plan over a simulated 30-day implementation period, performance metrics were re-evaluated for a subsequent 7-day post-TPM trial.

Table 10.1: Before vs After TPM Performance Comparison Table

PERFORMANCE INDICATOR (KPI)	BEFORE TPM (BASELINE)	AFTER TPM (PROJECTED)	ABSOLUTE CHANGE	RELATIVE IMPROVEMENT (%)
Planned Production Time	6,720 min	6,720 min	0 min	—

PERFORMANCE INDICATOR (KPI)	BEFORE TPM (BASELINE)	AFTER TPM (PROJECTED)	ABSOLUTE CHANGE	RELATIVE IMPROVEMENT (%)
Total Downtime	750 min	230 min	-520 min	**+69.33% Reduction**
Actual Operating Time	5,970 min	6,490 min	+520 min	**+8.71% Increase**
Availability (A)	88.84%	96.58%	+7.74%	**+8.71% Improvement**
Performance (P)	89.78%	95.50%	+5.72%	**+6.37% Improvement**
Quality (Q)	95.85%	99.20%	+3.35%	**+3.50% Improvement**
Overall Equipment Effectiveness (OEE)	**76.45%**	**91.50%**	**+15.05%**	**+19.69% Total OEE Gain**
Number of Breakdown Events	14 failures	4 failures	-10 failures	**+71.43% Failure Reduction**
MTBF (Mean Time Between Failures)	426.43 min (7.1 hrs)	1,622.50 min (27.0 hrs)	+1,196.07 min	**+280.48% MTBF Extension**
MTTR (Mean Time To Repair)	53.57 min	57.50 min	+3.93 min	*(Focused thorough repairs)*
Downtime Percentage	11.16%	3.42%	-7.74%	**+69.35% Downtime Reduction**

Figure 10.1: OEE & Key Metric Comparison (Before vs After TPM)



Projections Disclaimer: The "After TPM" metric values represent realistic industrial projections derived from systematic loss reduction models for learning purposes. They serve to illustrate theoretical gains achieved when TPM routines are strictly enforced.

SECTION 11: STANDARD OPERATING TPM DOCUMENTATION

Practical operational control requires clear, shop-floor ready documentation. Below are three standardized templates for conveyor management.

1. Autonomous Maintenance (AM) Daily Checklist

Equipment: Bakery Line Conveyor #04 • **Frequency:** Daily (Start of Shift) • **Operator Role:** Shift Operator

ITEM	INSPECTION TASK	STANDARD METHOD	ACCEPTABLE CRITERION	STATUS (OK/ NOK)	OPERATOR REMARKS
1	Sensors Lens Cleaning	Wipe with alcohol cloth	Lens clear, no grease film	[ ] OK	Cleaned at 06:00 AM

ITEM	INSPECTION TASK	STANDARD METHOD	ACCEPTABLE CRITERION	STATUS (OK/ NOK)	OPERATOR REMARKS
2	Belt Tracking Visual Check	Observe running belt edge	Centered within ±5mm of guide	[ ] OK	No rubbing against frame
3	Abnormal Noise & Vibration	Acoustic listening check	Smooth operation, no squeaking	[ ] OK	Bearings running smooth
4	Safety E-Stop Inspection	Press pull-cord switch	Conveyor stops instantly	[ ] OK	Tested & reset
5	Debris & Crumb Removal	Vacuum bottom pan	No crumb buildup under belt	[ ] OK	Cleared tray

2. Preventive Maintenance (PM) Schedule

Equipment: Food Conveyor System • **Frequency:** Weekly / Monthly • **Role:** Maintenance Technician

COMPONENT	MAINTENANCE TASK	FREQUENCY	INSPECTION STANDARD	TECHNICIAN ACTION
Drive Bearings	Re-greasing with H1 food grease	Weekly	2 pumps of grease gun per housing	Apply grease, wipe excess
Gearbox Unit	Oil level & clarity check	Monthly	Oil level at center sight glass mark	Top up synthetic food-grade oil
Conveyor Belt	Tension & splice join audit	Bi-Weekly	Belt sag between 15mm-20mm	Adjust screw jack tensioners
Drive Motor	Current draw & temp check	Monthly	Current ≤ 12.5A, Temp ≤ 65°C	Log thermal imaging reading

3. Machine Inspection Checklist (Quality & Safety Audit)

PARAMETER	TECHNICAL REQUIREMENT	MEASURED VALUE	COMPLIANCE	CORRECTIVE ACTION TAKEN
Belt Speed Sync	Target speed: 40.0 units/min	39.8 units/min	PASS	VFD speed setting calibrated.
Drive Chain Tension	Deflection: 10mm to 15mm	12mm deflection	PASS	Chain slack checked.
Guard Safety Interlock	Motor trips when guard removed	Trips immediately	PASS	Interlock functional.
Food Contact Hygiene	Zero microbial residue (ATP swab)	< 15 RLU (Clean)	PASS	Passed sanitation audit.

SECTION 12: PRACTICAL NUMERICAL QUESTIONS & STEP-BY-STEP SOLUTIONS

This section provides 15 structured engineering numerical problems designed to reinforce OEE and reliability calculations.

Q1. Availability Calculation

A conveyor system operates for a planned shift of 8 hours (480 minutes). During the shift, the conveyor suffers 45 minutes of unscheduled downtime. Calculate the Availability percentage.

Solution:

Planned Time = 480 min, Downtime = 45 min

Operating Time = 480 - 45 = 435 min

Availability = (435 / 480) × 100 = 90.625% ≈ 90.63%

Q2. Performance Calculation

A conveyor line operates for 400 minutes. The ideal cycle time per food container is 2.0 seconds (0.0333 min/unit). Total production output is 10,500 containers. Calculate the Performance percentage.

Solution:

Theoretical Time Required = 10,500 units × (2.0 / 60 min/unit) = 350 minutes

Performance = (350 min / 400 min) × 100 = 87.50%

Q3. Quality Calculation

During a production batch, 15,000 food packages were moved across the conveyor. Inspection identified 450 packages damaged by conveyor side-rail pinch points. Calculate the Quality percentage.

Solution:

Good Units = 15,000 - 450 = 14,550 units

Quality = (14,550 / 15,000) × 100 = 97.00%

Q4. Overall Equipment Effectiveness (OEE)

A packaging line achieves Availability = 92.0%, Performance = 88.0%, and Quality = 98.5%. Calculate the overall OEE.

Solution:

OEE = (0.920 × 0.880 × 0.985) × 100 = 79.7456% ≈ 79.75%

Q5. MTBF Calculation

Over a 30-day operating month, a conveyor runs for 420 hours in total and experiences 12 breakdown failures. Calculate the Mean Time Between Failures (MTBF) in hours.

Solution:

MTBF = Total Operating Hours / Number of Failures

MTBF = 420 hours / 12 failures = 35.00 hours/failure

Q6. MTTR Calculation

Refer to Q5: The total downtime incurred across the 12 failure events was 18 hours. Calculate the Mean Time To Repair (MTTR) in minutes.

Solution:

MTTR = Total Repair Time / Number of Repairs

MTTR = 18 hours / 12 repairs = 1.50 hours = 90.00 minutes/repair

Q7. Downtime Percentage

A facility plans 1,200 hours of conveyor operation in a quarter. Unscheduled maintenance accounts for 102 hours of downtime. Calculate the Downtime Percentage.

Solution:

Downtime % = (102 / 1,200) × 100 = 8.50%

Q8. Target Production Output Loss

A conveyor line with an ideal speed of 50 units/minute suffers 65 minutes of sensor breakdown downtime. How many units of lost production output occurred due to this downtime?

Solution:

Lost Production = Ideal Speed × Downtime

Lost Production = 50 units/min × 65 min = 3,250 units

Q9. OEE Percentage Improvement

Before implementing Kaizen, a plant's OEE was 68.0%. After implementing Autonomous Maintenance, OEE rose to 81.6%. Calculate the percentage improvement relative to baseline.

Solution:

$$\textit{Relative Improvement} = [(\textit{New OEE} - \textit{Baseline OEE}) / \textit{Baseline OEE}] \times 100$$

$$\textit{Improvement} = [(81.6 - 68.0) / 68.0] \times 100 = (13.6 / 68.0) \times 100 = 20.00\%$$

Q10. Finding Operating Time from Availability Target

A production manager targets Availability = 95.0% for a planned shift of 960 minutes. What is the maximum allowable downtime in minutes to meet this target?

Solution:

$$\textit{Required Operating Time} = 0.95 \times 960 = 912 \text{ minutes}$$

$$\textit{Maximum Allowable Downtime} = 960 - 912 = 48.00 \text{ minutes}$$

Q11. Speed Loss Analysis

A conveyor runs for 500 minutes and produces 12,000 units. The nameplate ideal speed is 30 units/minute. Calculate speed loss in units and Performance percentage.

Solution:

$$\textit{Theoretical Output} = 500 \text{ min} \times 30 \text{ units/min} = 15{,}000 \text{ units}$$

$$\textit{Speed Loss Output} = 15{,}000 - 12{,}000 = 3{,}000 \text{ units}$$

$$\textit{Performance} = (12{,}000 / 15{,}000) \times 100 = 80.00\%$$

Q12. Multi-Shift MTBF Comparison

Shift A operates 400 hours with 2 failures. Shift B operates 400 hours with 8 failures. Calculate MTBF for both shifts and state which shift has higher reliability.

Solution:

$$\textit{Shift A MTBF} = 400 / 2 = 200 \text{ hours/failure}$$

$$\textit{Shift B MTBF} = 400 / 8 = 50 \text{ hours/failure}$$

Conclusion: Shift A has 4 times higher reliability than Shift B.

Q13. Financial Loss of Downtime

A conveyor breakdown halts production for 2.5 hours. The line produces food items with a profit margin of $1.50 per unit, and the standard production rate is 1,200 units/hour. Calculate total direct profit loss.

Solution:

$$\textit{Total Lost Units} = 2.5 \text{ hours} \times 1{,}200 \text{ units/hr} = 3{,}000 \text{ units}$$

$$\textit{Financial Loss} = 3{,}000 \text{ units} \times \$1.50/\text{unit} = \$4{,}500.00$$

Q14. Quality Scrap Rate Calculation

If a conveyor system has a Quality rating of 96.5% and produced 28,000 total units, how many units were scrapped or sent for rework?

Solution:

$$\textit{Scrap Rate \%} = 100\% - 96.5\% = 3.5\%$$

$$\textit{Defective/Scrap Units} = 28{,}000 \times 0.035 = 980 \text{ units}$$

Q15. OEE Bottleneck Identification

A machine has A = 95%, P = 70%, Q = 99%. Identify the primary bottleneck factor reducing OEE and suggest a TPM pillar to fix it.

Solution:

$$OEE = 0.95 \times 0.70 \times 0.99 = 65.835\%$$

Primary Bottleneck: Performance (70%) due to micro-stoppages or reduced speed operation. **Recommended Pillar:** Focused Improvement (Kobetsu Kaizen) to eliminate minor idling losses.

SECTION 13: VIVA QUESTIONS AND CONCISE ANSWERS

1. **What is the fundamental philosophy of Total Productive Maintenance (TPM)?**
   Answer: To maximize equipment efficiency and eliminate all downtime, defects, and safety accidents through plant-wide employee involvement and proactive daily maintenance.
2. **Name the eight pillars of TPM.**
   Answer: 1. Autonomous Maintenance, 2. Planned Maintenance, 3. Focused Improvement (Kaizen), 4. Quality Maintenance, 5. Early Equipment Management, 6. Training & Education, 7. Safety Health & Environment, 8. Office TPM.
3. **What is Autonomous Maintenance (Jishu Hozen)?**
   Answer: A practice where machine operators perform routine cleaning, lubrication, visual inspection, and basic adjustments on their own equipment rather than relying solely on technicians.
4. **What are the three factors used to calculate OEE?**
   Answer: Availability (downtime loss), Performance (speed loss), and Quality (defect loss).
5. **Write the mathematical formula for OEE.**
   Answer: $ ext{OEE} = ext{Availability} imes ext{Performance} imes ext{Quality}$.
6. **What is the difference between MTBF and MTTR?**
   Answer: MTBF measures machine reliability (average operating time between breakdowns), while MTTR measures repair efficiency (average time required to fix a breakdown).
7. **What constitutes a "world-class" OEE benchmark?**
   Answer: An overall OEE of 85% or higher, typically composed of Availability ≥ 90%, Performance ≥ 95%, and Quality ≥ 99%.
8. **Explain the difference between Preventive and Predictive Maintenance.**
   Answer: Preventive maintenance is scheduled based on fixed time/cycles regardless of condition. Predictive maintenance monitors real-time condition (vibration, temp) and intervenes only when degradation occurs.
9. **How does belt misalignment affect conveyor effectiveness?**
   Answer: Misalignment causes belt edge friction, frame wear, motor thermal overload due to friction drag, and product tipping, severely reducing Availability and Quality.
10. **What is the 80/20 rule in Pareto Analysis?**
    Answer: It states that roughly 80% of equipment downtime or failure problems originate from 20% of the root causes.
11. **Why is food-grade lubricant (H1) mandatory in conveyor maintenance?**
    Answer: H1 lubricants are non-toxic and food-safe, ensuring public health safety in case of incidental contact with food products on the belt.
12. **What is Kaizen in the context of TPM?**
    Answer: Continuous, small-step incremental improvements aimed at eliminating specific equipment losses and operational inefficiencies.
13. **How do dirty optical sensors degrade conveyor Performance?**
    Answer: Lens fouling causes false optical triggers, resulting in frequent micro-stoppages and artificial speed throttling by automated VFD controllers.
14. **What is the concept of "Three Zeros" in TPM?**
    Answer: Zero Breakdowns, Zero Quality Defects, and Zero Workplace Accidents.
15. **Why must operating conditions be considered when interpreting OEE benchmarks?**
    Answer: Benchmarks vary based on process complexity, batch changeovers, and equipment age; universal guarantees do not apply to custom manufacturing lines.

SECTION 14: CONCLUSION

In this experimental study, the principles of Total Productive Maintenance (TPM) were applied to analyze and improve an automated food processing conveyor system. A simulated 7-day operational baseline dataset was collected, revealing an initial Overall Equipment Effectiveness (OEE) of **76.45%** (Availability: 88.84%, Performance: 89.78%, Quality: 95.85%). Reliability analysis indicated an MTBF of 426.43 minutes and an MTTR of 53.57 minutes, with total downtime accounting for 11.16% of planned production time.

Downtime Pareto analysis established that belt misalignment, sensor fouling, and bearing lubrication issues constituted 76% of total unscheduled downtime. Implementation of Autonomous Maintenance routines, laser-aligned belt tracking, automated H1 lubrication, and operator single-point lessons projected a significant performance enhancement: OEE increased to **91.50%**, downtime decreased by 69.33%, and MTBF extended to 1,622.50 minutes.

This experiment confirms that TPM provides a systematic, data-driven engineering framework for eliminating equipment losses, maximizing process throughput, and fostering a culture of operational excellence in modern manufacturing industries.

SECTION 15: QUICK FORMULA SHEET

NO.	PERFORMANCE METRIC	STANDARD ENGINEERING FORMULA	KEY SYMBOLS & DEFINITIONS
1	**Availability (A)**	$A = (Operating\ Time / Planned\ Prod.\ Time) \times 100$	$Operating\ Time = Planned\ Time - Downtime$
2	**Performance (P)**	$P = [(Ideal\ Cycle\ Time \times Total\ Output) / Operating\ Time] \times 100$	$Ideal\ Cycle\ Time$ in min/unit; $Total\ Output$ in units
3	**Quality (Q)**	$Q = (Good\ Units / Total\ Units\ Produced) \times 100$	$Good\ Units = Total\ Units - Defect\ Units$
4	**Overall Equipment Effectiveness**	$OEE = (A \times P \times Q) / 10,000$	Expressed as overall percentage (%)
5	**Mean Time Between Failures**	$MTBF = Total\ Operating\ Time / Total\ Failures$	Units: Operating hours (or minutes) per failure
6	**Mean Time To Repair**	$MTTR = Total\ Maintenance\ Downtime / Total\ Repairs$	Units: Minutes (or hours) per repair event
7	**Downtime Percentage**	$Downtime\ \% = (Total\ Downtime / Planned\ Time) \times 100$	Unscheduled downtime fraction (%)
8	**Percentage Improvement**	$\%\ Imp = [(New\ Value - Base\ Value) / Base\ Value] \times 100$	Relative KPI growth measurement (%)